# Biology

# Class 10 (English Medium)

# Solved Exercises — Sindh Textbook Board

9 lessons — 209 questions

## گئسن جي مٿا سٿا

1. The life process in which gaseous exchange takes place is:

(a) photosynthesis

(b) respiration

✓ (c) both of these

(d) none of these

2. Plants carry out gaseous exchange through _____:

(a) root

(b) stomata

(c) stem

✓ (d) all of these

3. Each stoma is surrounded by:

(a) one guard cell

✓ (b) two guard cells

(c) three guard cells

(d) four guard cells

4. The respiratory surface has the following properties:

(a) thin and moist

(b) permeable

(c) very large

✓ (d) all of these

5. The intake of air involves:

(a) contraction of the intercostal muscles

(b) contraction of the diaphragm

(c) inward movement of the ribs

✓ (d) both a and b

6. The larynx is present in the:

(a) pharynx

✓ (b) trachea

(c) bronchi

(d) bronchioles

7. The respiratory surface in humans is the:

(a) nose

(b) bronchioles

✓ (c) alveoli

(d) trachea

8. The breathing rate increases because the level of _____ in the blood rises:

✓ (a) carbon dioxide

(b) oxygen

(c) carbon dioxide in the lungs

(d) oxygen in the lungs

9. The disorder linked to damage of the alveoli is:

(a) bronchitis

(b) lung cancer

(c) asthma

✓ (d) emphysema

10. The disorder linked to narrowing of the air passage is:

(a) bronchitis

(b) lung cancer

✓ (c) asthma

(d) emphysema

Question: 11. Why are stomata generally open during the daytime?

Answer: During the day carbon dioxide is needed for photosynthesis, so

the stomata stay open so that carbon dioxide can enter and oxygen can leave.

Question: 12. Which parts of plants take in carbon dioxide and release oxygen?

Answer: Mainly the leaves take in carbon dioxide and release oxygen through the stomata; the stem exchanges gases through lenticels and the root also exchanges gases.

Question: 13. Why is breathing through the nose better than breathing through the mouth?

Answer: The nose has hairs and mucus which trap dust and germs and clean the air, and warm and moisten it; the mouth does not provide this protection.

Question: 14. Distinguish between breathing, gaseous exchange and respiration.

Answer: Breathing (ventilation) is the movement of air in and out; gaseous exchange is the swapping of gases between the body and the environment; respiration is the chemical process in cells in which food is broken down to release energy.

Question: 15. Why does the breathing rate rise during or after exercise?

Answer: During exercise the muscles need more energy, which needs more oxygen and produces more carbon dioxide to be removed, so breathing becomes faster.

Question: 16. What is oxygen debt?

Answer: During hard exercise the muscles work without enough oxygen and lactic acid is formed; the extra oxygen needed afterwards to remove that lactic acid is called oxygen debt.

Question: 17. What is the difference between inhaled and exhaled air?

Answer: Inhaled air has more oxygen and less carbon dioxide; exhaled air

has less oxygen and more carbon dioxide, and it is warm and moist.

Question: 18. What is lung cancer?

Answer: Lung cancer is the uncontrolled growth of cells in the lungs; it is mostly caused by smoking and makes breathing difficult.

Question: 19. How can an asthma patient be protected?

Answer: An asthma patient can be protected by avoiding dust, smoke, cold air and other triggers, taking medicine on time, living in a clean environment and following the doctor's instructions.

Question: 20. Name animals that exchange gases through the body surface.

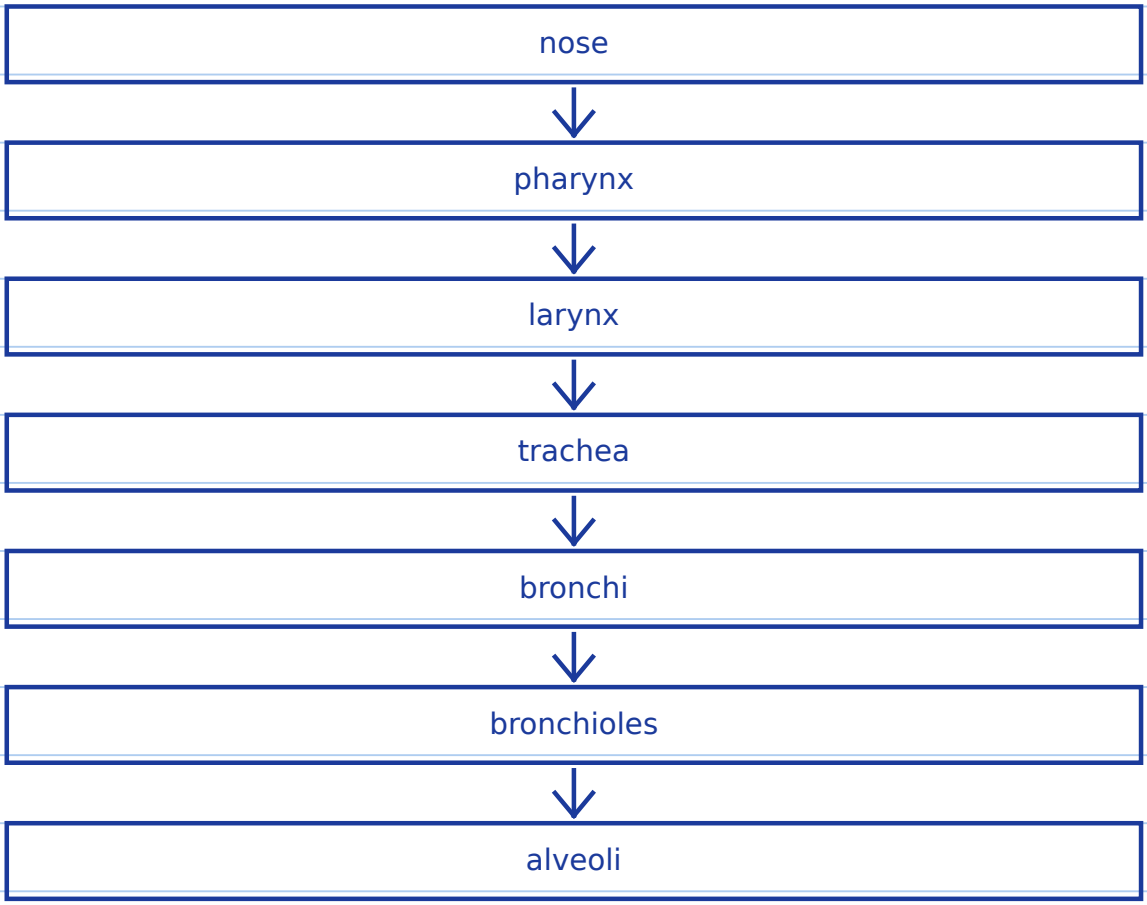
Answer: The earthworm, the frog (through its skin) and single-celled organisms like Amoeba exchange gases through the body surface.

Question: 21. What steps should be taken to avoid respiratory disorders?

Answer: Avoid smoking, live in clean air, stay away from dust and smoke, exercise regularly, eat a balanced diet, get vaccinated, and seek timely treatment if illness occurs.

Question: 22. Describe the human respiratory system with a labelled diagram.

Answer: The human respiratory system consists of the nose, pharynx, larynx, trachea, bronchi, bronchioles and alveoli; air enters through the nose, passes through the pharynx and larynx into the trachea, then through the bronchi and bronchioles to the alveoli, where gaseous exchange takes place. (See the labelled diagram in the book.)




Labelled diagram — schematic; parts as named in the answer. See the book diagram for exact form.

Question: 23. Describe the experiment that proves carbon dioxide is released during respiration.

Answer: Place germinating seeds in a flask and pass the air coming from it through lime water; the lime water turns milky, which proves that carbon dioxide is released during respiration.

Question: 24. Describe ventilation (inhalation and exhalation) in humans in detail.

Answer: In inhalation (inspiration) the intercostal muscles and the diaphragm contract, the chest capacity increases and air rushes in; in exhalation (expiration) the muscles relax, the chest capacity decreases and air is pushed out.

Question: 25. Why is smoking harmful to health, and which respiratory

disorders are connected with it?

Answer: Cigarette smoke contains poisonous substances like nicotine, tar and carbon monoxide which damage the lungs and the air passage; it can cause bronchitis, emphysema, asthma and lung cancer.

## ہومیوسٹیسس

1. The internal condition of an organism is called:

✓ (a) homeostasis

(b) internal environment

(c) internal biological process

(d) feedback

2. The combined biological processes that keep the internal environment stable are called:

(a) negative feedback

(b) positive feedback

(c) osmoregulation

✓ (d) homeostasis

3. The removal of extra water from the body in the form of vapour is called:

(a) excretion

(b) guttation

(c) respiration

✓ (d) transpiration

4. Plants that grow on the seashore are:

(a) xerophytes

✓ (b) halophytes

(c) epiphytes

(d) hydrophytes

5. The largest organ of the human body is considered to be the:

✓ (a) skin

(b) digestive tract

(c) liver

(d) brain

6. Keeping the body temperature steady at a set level is called:

(a) homeotherm

✓ (b) thermoregulation

(c) osmoregulation

(d) heterotherm

7. The kidney is enclosed in a membrane called the:

(a) pericardium

✓ (b) peritoneum

(c) pleural membrane

(d) plumule

8. The layer of the skin containing a network of blood vessels is called the:

(a) epidermis

✓ (b) dermis

(c) hypodermis

(d) endodermis

9. In the nephron, selective reabsorption takes place in the:

(a) glomerulus

(b) Malpighian body

✓ (c) convoluted tubule

(d) loop of Henle

10. The ADH hormone is secreted by which organ of the body:

✓ (a) pituitary gland

(b) kidney

(c) liver

(d) spleen

Question: 11. Why does the body need homeostasis?

Answer: Homeostasis is needed to keep the body's internal environment (temperature, water, salts and pH) steady so that the cells and organs can work properly.

Question: 12. By which ways do plants remove extra water?

Answer: Plants remove extra water by transpiration (as vapour through the leaves), by guttation (as drops from the leaf edges) and by excretion.

Question: 13. How can plants survive in salty water?

Answer: Halophyte plants survive in salty water by storing salt in their cells, removing extra salt, and keeping their water balance steady.

Question: 14. Why is the skin considered an excretory organ?

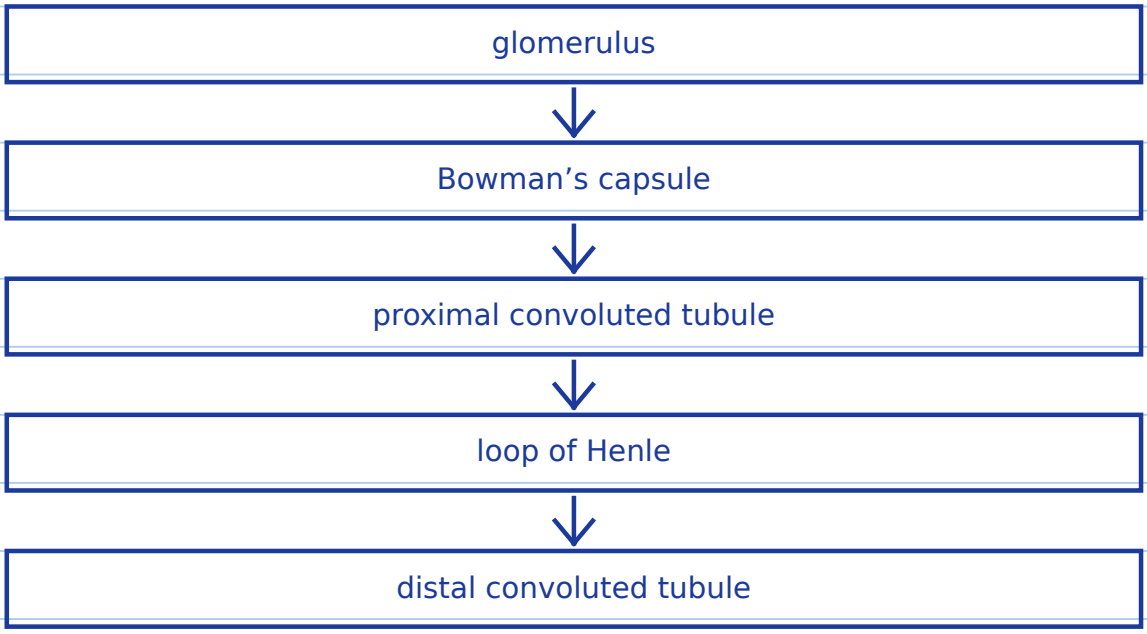
Answer: The skin removes extra water, salts and waste substances like urea from the body through sweat, so it is considered an excretory organ.

Question: 15. What are the parts of the structure present in the kidney and what work do they do?

Answer: The kidney contains the cortex (outer part), the medulla (inner part), the pelvis (urine-collecting part), the nephrons (units that filter the blood and make urine) and the ureter (the tube carrying urine to the bladder).

Question: 16. Describe the structure of the nephron with a neat labelled diagram.

Answer: The nephron consists of the glomerulus (a ball of blood capillaries), Bowman's capsule, the proximal convoluted tubule, the loop of Henle and the distal convoluted tubule; blood is filtered in the glomerulus and useful substances are reabsorbed in the tubule. (See the labelled diagram in the book.)




Labelled diagram — schematic; parts as named in the answer. See the book diagram for exact form.

Question: 17. How does human skin keep the body temperature steady in heat?

Answer: In heat the blood vessels of the skin widen, more sweat is produced, and the evaporation of sweat cools the body, keeping the temperature steady.

Question: 18. Why is dialysis needed?

Answer: When the kidneys fail and cannot remove waste substances and extra water from the blood, dialysis is needed to clean the blood with a machine.

Question: 19. What is the function of tubular secretion and reabsorption in the tubules?

Answer: Reabsorption returns useful substances like water, glucose and salts to the blood; tubular secretion moves extra and harmful substances from the blood into the tubule so they leave in the urine.

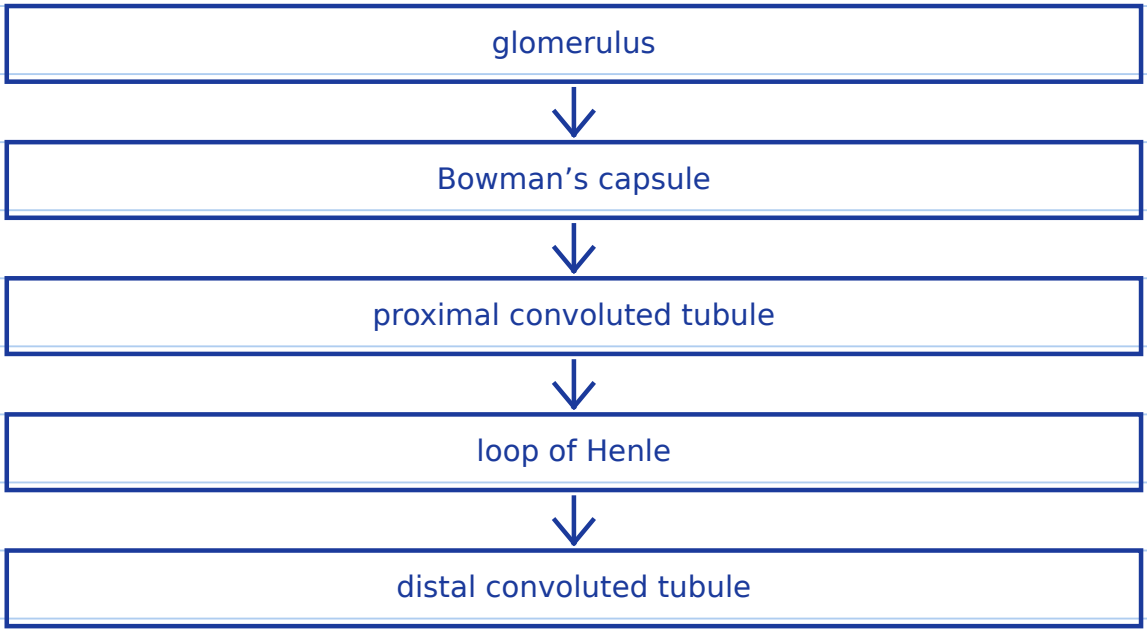
Question: 20. How does the skin work as a thermoregulatory organ?

Answer: In heat the skin widens the blood vessels and increases sweating

to cool the body, and in cold it narrows the blood vessels and reduces sweating to conserve heat; in this way it keeps the body temperature steady.

Question: 21. Describe the structure of the nephron with a diagram.

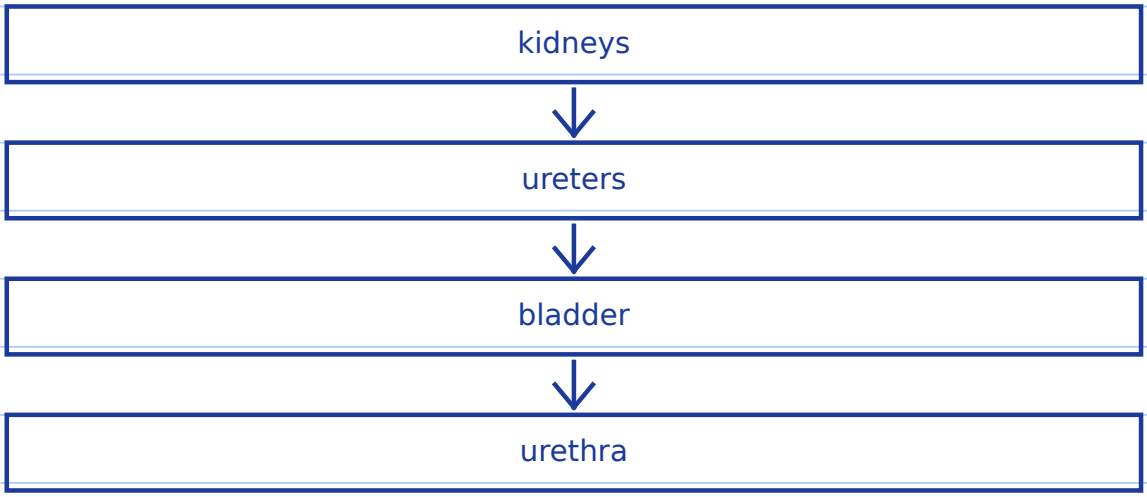
Answer: The nephron is the unit of the kidney; it includes the glomerulus, Bowman's capsule, the proximal convoluted tubule, the loop of Henle, the distal convoluted tubule and the collecting duct. Blood is filtered in the glomerulus, selective reabsorption and secretion take place in the tubule, and urine collects in the collecting duct. (See the diagram in the book.)




Labelled diagram — schematic; parts as named in the answer. See the book diagram for exact form.

Question: 22. Describe the human urinary system with a diagram.

Answer: The human urinary system consists of two kidneys, two ureters, one bladder and one urethra; the kidneys make urine, the ureters carry it to the bladder, the bladder stores it, and it leaves through the urethra. (See the diagram in the book.)




Labelled diagram — schematic; parts as named in the answer. See the book diagram for exact form.

Question: 23. Describe the role of the blood-vessel network in the nephron.

Answer: The glomerulus (a ball of blood capillaries) filters the blood; the network of blood vessels around the tubule takes reabsorbed substances back into the blood and supplies substances for secretion; thus the blood network plays the central role in making urine.

Question: 24. Describe kidney diseases and their treatment.

Answer: Kidney diseases include kidney stones, nephritis (inflammation of the kidney), urinary tract infection and kidney failure; treatment includes medicines, breaking up stones (lithotripsy) or surgery, dialysis, and kidney transplant.

# هر آهنگي

1. Movements performed due to a change in the environment are called:

(a) stimulus

✓ (b) response

(c) both a and b

(d) none of these

2. A stimulus is perceived through the:

✓ (a) receptor

(b) effector

(c) neuron

(d) all of these

3. Rapid communication through electrochemical signals is:

✓ (a) nervous

(b) chemical

(c) mechanical

(d) all

4. A chemical released from one cell that signals another cell at a distance or is sent through the blood is a:

(a) neurotransmitter

(b) enzyme

✓ (c) hormone

(d) all

5. The method of coordination in plants is:

(a) nervous

✓ (b) chemical

(c) mechanical

(d) both a and b

6. The part of the brain connected with reasoning is the:

(a) forebrain

(b) cerebrum

(c) cortex

✓ (d) frontal lobe

7. The part of the brain that produces balance and precision in movement is the:

(a) cerebrum

✓ (b) cerebellum

(c) thalamus

(d) medulla oblongata

8. The part of the nervous system controlling survival actions is the:

(a) CNS

(b) PNS

(c) somatic

✓ (d) autonomic

9. A reflex action consists of at most how many neurons:

(a) 1

(b) 2

✓ (c) 3

(d) many

10. What type of lens is present in the human eye:

✓ (a) convex

(b) concave

(c) both

(d) none

11. Focusing the eye so that a near object gives a clear image is called:

(a) vision

✓ (b) accommodation

(c) focus

(d) all

12. Which is needed for correct vision:

✓ (a) vitamin A

(b) vitamin B

(c) vitamin C

(d) vitamin D

13. What can a colour-blind person not see:

(a) everything

✓ (b) green colour

(c) white colour

(d) black colour

14. Photoreceptor cells are present in the:

✓ (a) retina

(b) cochlea

(c) skin

(d) nose

15. Gonads are target organs of:

(a) FSH

(b) LH

✓ (c) both

(d) none

Question: 16. What is meant by feedback system?

Answer: A feedback system is one in which the result of an action is reported back to the system so that the action can be increased or decreased and balance is maintained.

Question: 17. Why does nervous coordination work faster than chemical coordination?

Answer: Nervous coordination travels directly and fast through electrochemical signals, while in chemical coordination hormones travel slowly through the blood.

Question: 18. Which of the two coordination systems is better?

Answer: Both systems are necessary in their place; the nervous system acts fast and briefly, while the chemical (endocrine) system acts slowly but for a long time, so together they work best.

Question: 19. How does a reflex action work?

Answer: The stimulus is felt by a receptor, the signal travels through a sensory neuron to the spinal cord, from there through a motor neuron to the effector (muscle), and the muscle moves at once, without the brain having to think.

Question: 20. Why can a colour-blind person not distinguish certain colours?

Answer: In a colour-blind person some cone cells in the retina are defective or absent, so he cannot distinguish those colours.

Question: 21. Why does the thyroid gland swell and what is it called?

Answer: Due to iodine deficiency the thyroid gland swells, which is called goitre.

Question: 22. What work do the islets of Langerhans perform?

Answer: The islets of Langerhans in the pancreas make the hormones insulin and glucagon, which control the blood glucose level.

Question: 23. What are emergency hormones and what are they called?

Answer: The emergency hormones are adrenaline and noradrenaline, called the hormones of the adrenal glands; they prepare the body in times of danger.

Question: 24. What are the possible causes of paralysis?

Answer: Possible causes of paralysis are bursting or blockage of a blood vessel in the brain, brain injury, high blood pressure and nerve damage.

Question: 25. What is epilepsy?

Answer: Epilepsy is a disorder of the brain in which sudden abnormal electrical activity in the brain causes fits of jerking movements and unconsciousness.

Question: 26. What is the difference between chemical and nervous coordination?

Answer: Nervous coordination is fast and short-lived through electrochemical signals; chemical coordination is slow and long-lasting through hormones.

Question: 27. Describe the structure of the human brain in detail.

Answer: The human brain includes the cerebrum (the largest part, thinking and memory), the cerebellum (balance and precision of movement) and the brain stem (medulla, pons, midbrain — basic actions like breathing and heartbeat); it is covered by three meninges membranes and has ventricles inside.

Question: 28. Write a detailed note on the human eye.

Answer: The human eye has the sclera (white outer coat), cornea (transparent front part), iris (coloured part), pupil (opening), lens (convex), vitreous and retina (rod and cone cells); light passes through the cornea and lens to form an image on the retina, and the signal travels to the brain through the optic nerve.

Question: 29. Why do people differ in intelligence though the parts of the brain are alike?

Answer: Intelligence depends not only on the parts of the brain but on the number of neurons, the connections between them, the growth of the

brain, education and environment; differences in these cause differences in intelligence.

Question: 30. What is the endocrine system, and how do these glands keep the blood glucose level steady?

Answer: The endocrine system is the system of glands that release hormones into the blood; when blood glucose rises the pancreas releases insulin which lowers it, and when glucose falls it releases glucagon which raises it, so the level stays steady.

Question: 31. What is diabetes and how can it be prevented?

Answer: Diabetes (sugar) is the disease in which blood glucose rises due to lack of insulin or its failure to act; it can be prevented by a balanced diet, exercise, keeping weight under control and regular check-ups.

# سہارو حرکت

1. The restlessness produced due to a stimulus is removed by:

(a) tropism

✓ (b) movement

(c) locomotion

(d) arthritis

2. The framework that gives a structure a firm shape is the:

(a) architecture

(b) bone

✓ (c) skeleton

(d) cartilage

3. The cell of cartilage is called:

(a) osteocyte

(b) osteoclast

✓ (c) chondrocyte

(d) osteoblast

4. The sacral vertebrae join with the:

✓ (a) pelvic girdle

(b) pectoral girdle

(c) scapula

(d) acetabulum of the lumbar region

5. The big muscle of the thigh is the:

(a) biceps

(b) triceps

(d) pentaceps

6. The strong band joining bone to bone at a joint is the:

(a) ligament

(b) tendon

(c) biceps

(d) triceps

7. Softening of children's bones due to vitamin D deficiency is:

(a) osteoporosis

(b) osteoarthritis

✓ (c) rickets

(d) rheumatic fever

8. The movement that straightens a joint is called:

(a) ligament

(b) skeleton

(c) flexion

✓ (d) extension

9. The pair of muscles working in opposite directions on a bone is the:

✓ (a) antagonist

(b) abductor

(c) adductor

(d) flexor

Question: 10. What are the most important functions of the skeleton?

Answer: The skeleton gives the body shape, protects the internal organs, helps movement, makes blood cells, and stores minerals like calcium.

Question: 11. What is the difference between bone and cartilage?

Answer: Bone is hard and strong and has blood vessels; cartilage is soft and flexible and has no blood vessels.

Question: 12. Name the cartilages found in different organs.

Answer: Cartilage is found in the ears, the tip of the nose, the rings of the trachea, the surfaces of joints and the ends of the ribs.

✓

Question: 13. List the names and number of bones in the legs/feet.

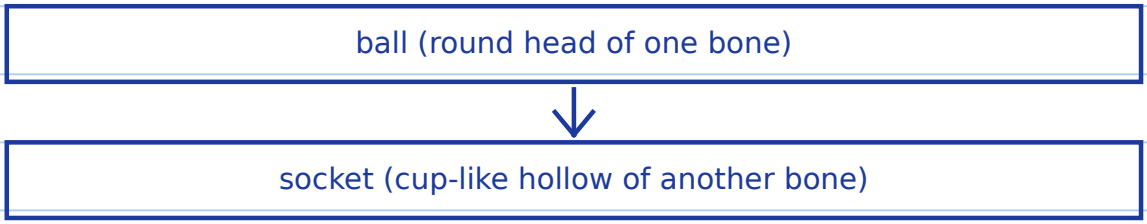
Answer: Each leg has the femur (thigh bone), tibia and fibula (shin bones), patella (knee cap), tarsals (7 ankle bones), metatarsals (5 bones) and phalanges (14 toe bones).

Question: 14. Where are immovable joints found in humans?

Answer: Immovable (fixed) joints are found between the bones of the skull and between the teeth and the jaws.

Question: 15. Draw a neat labelled diagram of the ball-and-socket joint, naming each part.

Answer: In a ball-and-socket joint the round head (ball) of one bone fits into the cup-like hollow (socket) of another bone; examples: the shoulder and hip joints. (See the labelled diagram in the book.)




Labelled diagram — schematic; parts as named in the answer. See the book diagram for exact form.

Question: 16. What is rickets and what is its cause?

Answer: Rickets is the disease of children in which the bones become soft and weak; its cause is deficiency of vitamin D, calcium or phosphorus.

Question: 17. What are the types of joints in the human body?

Answer: The human body has fixed (immovable), cartilaginous (slightly movable) and synovial (freely movable) joints; synovial joints include hinge, ball-and-socket, pivot and gliding joints.

Question: 18. What is the difference between ligament and tendon?

Answer: A ligament joins bone to bone; a tendon joins muscle to bone.

Question: 19. What is meant by biceps and triceps muscles?

Answer: The biceps is the muscle with two heads which bends the arm; the triceps is the muscle with three heads which straightens the arm; the two form an antagonist pair.

Question: 20. Describe that the skeleton is a strong system.

Answer: The skeleton is a strong framework of 206 bones which gives the body shape and support, protects delicate organs like the brain, heart and lungs, produces movement together with the muscles, and makes blood cells; its strength comes from the hard structure of bones.

Question: 21. Describe flexion and extension in humans in detail.

Answer: In flexion a muscle contracts and bends a joint (like bending the arm); in extension the opposite muscle contracts and straightens the joint (like straightening the arm); both happen through antagonist pairs of muscles.

Question: 22. Describe the types of joints in detail.

Answer: Fixed joints are immovable (skull); cartilaginous joints move slightly (between ribs and vertebrae); synovial joints move freely — hinge joints (knee, elbow) in one direction, ball-and-socket joints (shoulder, hip) in all directions, pivot joints (neck) in rotation, and gliding joints (wrist, ankle) in sliding movement.

# تولید

1. The process necessary for the survival of organisms and the continuation of their life is:

(a) care

(b) breathing

✓ (c) reproduction

(d) excretion

2. The type of reproduction necessary for evolution is:

(a) vegetative propagation

(b) budding

✓ (c) sexual reproduction

(d) cloning

3. Single-celled organisms reproduce asexually without fusion by means of:

(a) holes

✓ (b) spores

(c) gametes

(d) zygotes

4. An example of a stem that obtains little food lying on the ground and reproduces vegetatively:

✓ (a) mint

(b) ginger

(c) onion

(d) stone

5. The stem of a plant arising from the root of the parent plant is the:

(a) bulb

(b) rhizome

✓ (c) sucker

(d) runner

6. The method of seed production without male-female fusion is:

(a) parthenocarpy

✓ (b) apomixis

(c) parthenogenesis

(d) scion

7. In angiosperms, the female gametophyte of plants is called the:

✓ (a) embryo sac

(b) ovule

(c) ovary

(d) carpel

8. In angiosperms, from the 3N zygote comes the:

(a) seed coat

(b) cotyledon

(c) embryo

✓ (d) endosperm

9. The male gonad is located in the _____:

(a) vas deferens

(b) epididymis

✓ (c) scrotal sac

(d) penis

10. Female gametes are released from the:

(a) epididymis

(b) fallopian tube

✓ (c) ovary

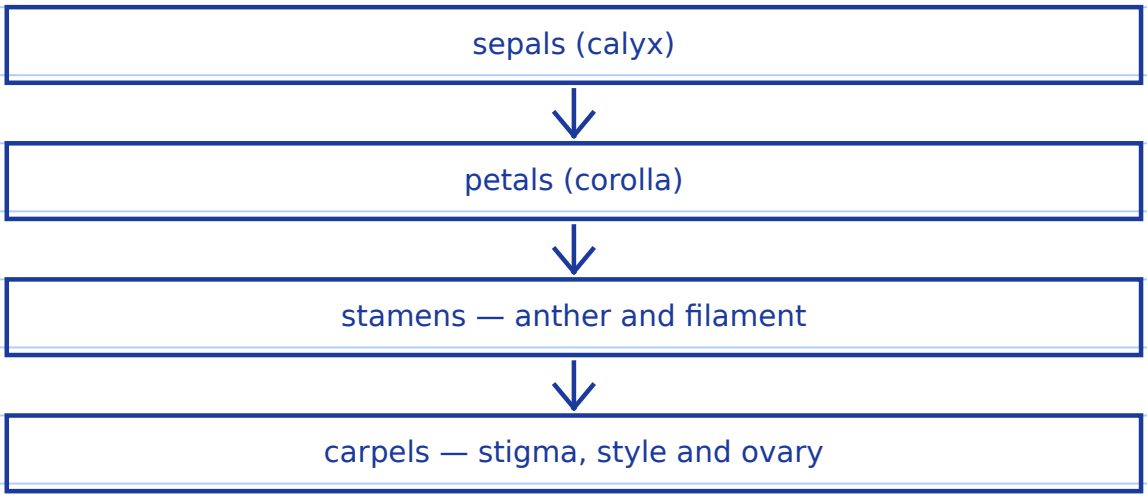
(d) uterus

Question: 11. State the difference between asexual reproduction, sexual reproduction and vegetative propagation.

Answer: In asexual reproduction a new organism is formed from a single parent without gametes; in sexual reproduction a new organism is formed by the fusion of male and female gametes; in vegetative propagation a new plant is formed from a part of the plant (stem, root, leaf).

Question: 12. Draw a neat labelled diagram of an angiosperm flower, naming each part.

Answer: A flower has sepals (calyx), petals (corolla), stamens (male part: anther and filament) and carpels (female part: stigma, style and ovary). (See the labelled diagram in the book.)




Labelled diagram — schematic; parts as named in the answer. See the book diagram for exact form.

Question: 13. How does a new plant arise from a single plant?

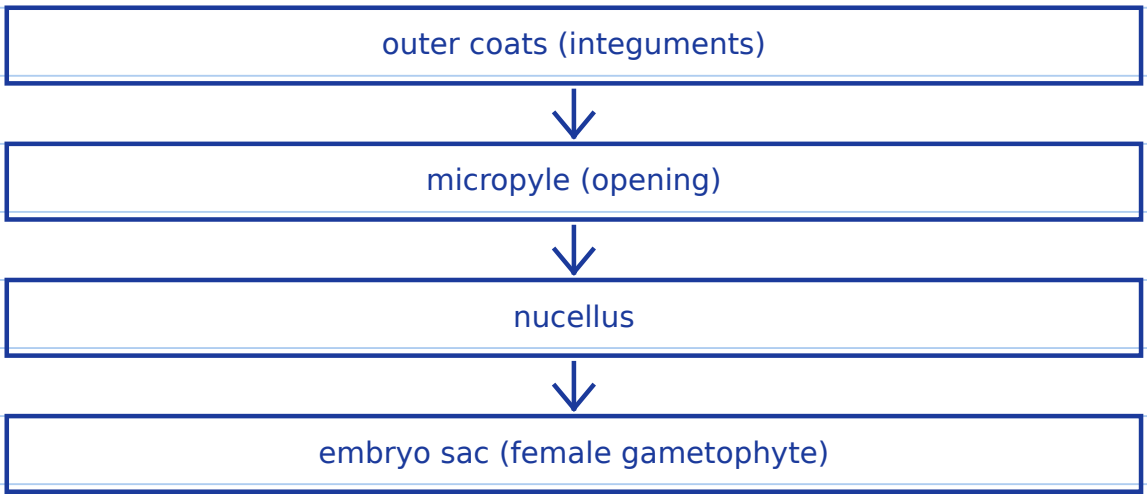
Answer: A new plant arises from a single plant by vegetative propagation — a new plant is formed from a part of the stem, root or leaf.

Question: 14. How does a new plant obtain food?

Answer: At first a new plant gets food stored in the parent plant or the seed, then it takes in water and minerals through the roots and makes its own food by photosynthesis in the leaves.

Question: 15. Draw a neat diagram of the ovule, naming each part.

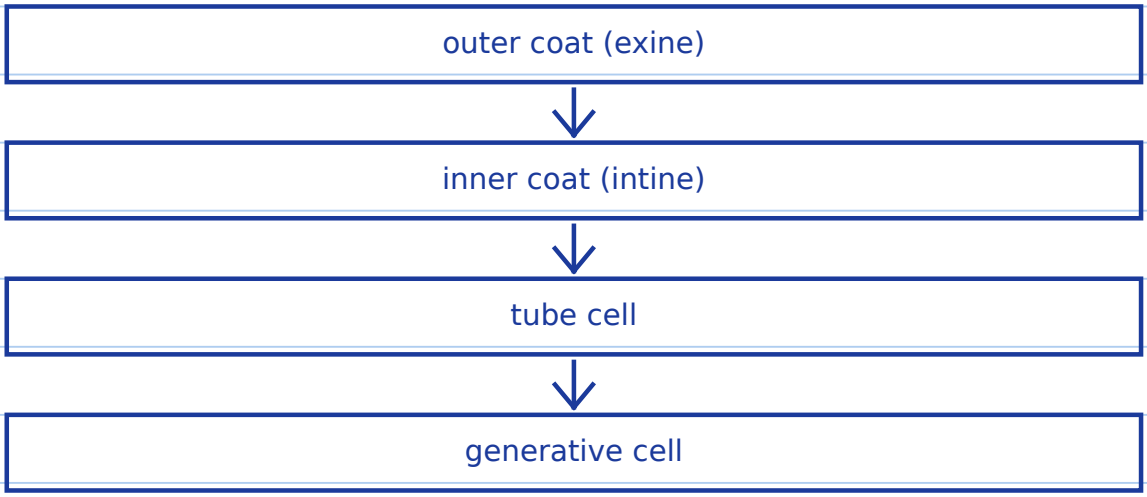
Answer: The ovule has outer coats (integuments), the micropyle (opening), the nucellus and the embryo sac (female gametophyte). (See the diagram in the book.)




Labelled diagram — schematic; parts as named in the answer. See the book diagram for exact form.

Question: 16. Draw a neat diagram of the male gametophyte of angiosperms, naming its different parts.

Answer: The male gametophyte (pollen grain) has an outer coat (exine), an inner coat (intine), a tube cell and a generative cell; the generative cell forms two male gametes. (See the diagram in the book.)




Labelled diagram — schematic; parts as named in the answer. See the book diagram for exact form.

Question: 17. List the organs of the male reproductive system and state the glands.

Answer: The male reproductive system includes the testes (gonads), epididymis, vas deferens, seminal vesicles, prostate gland, Cowper's glands and penis.

Question: 18. What is STD?

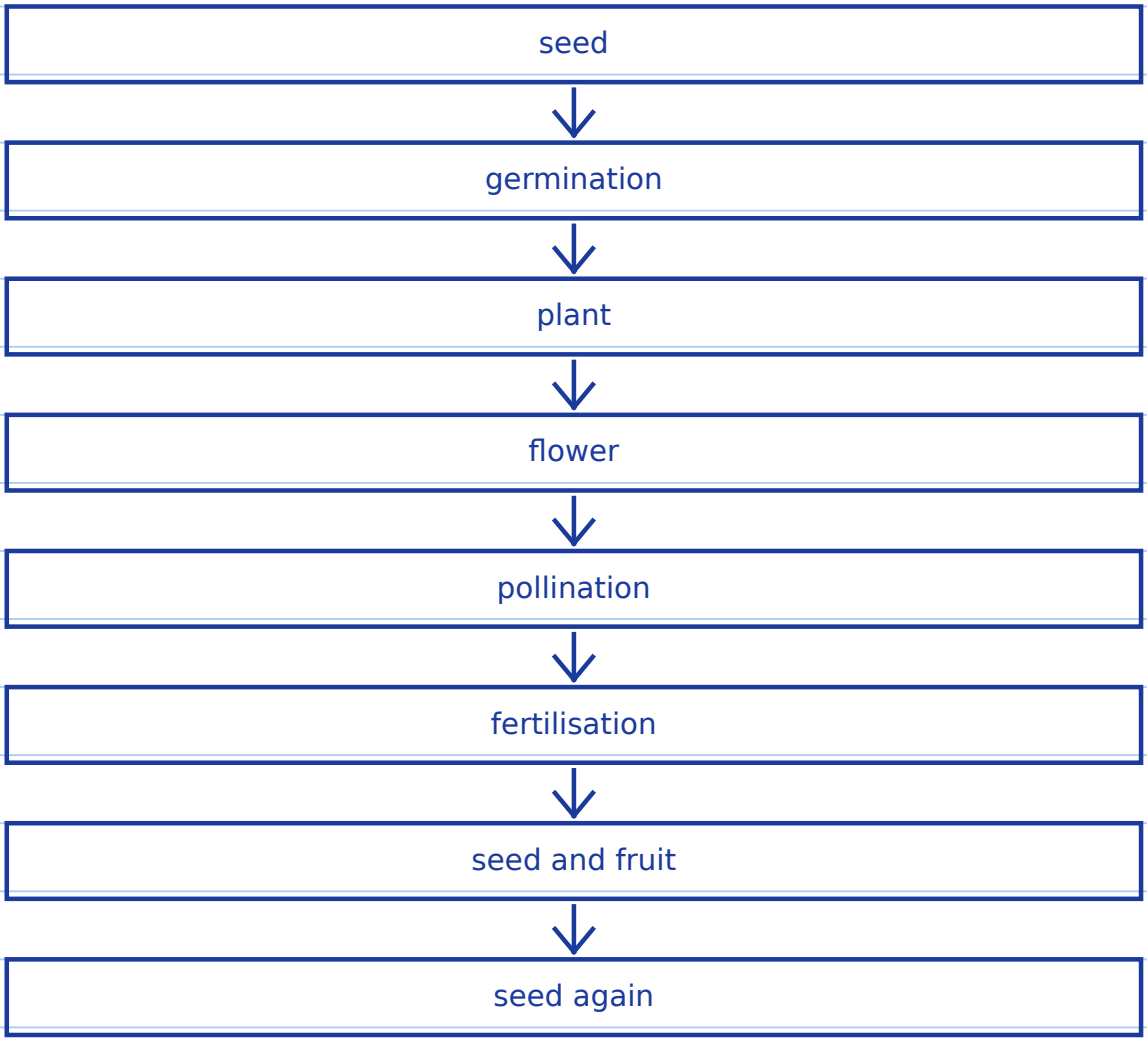
Answer: STD (sexually transmitted diseases) are diseases spread from one person to another through sexual contact, such as gonorrhoea, syphilis and AIDS.

Question: 19. Why is control of human population needed for a successful society?

Answer: Growing population increases pressure on food, water, housing, education, health and jobs; population control makes resources enough for everyone and the society successful.

Question: 20. Draw the life cycle of plants in angiosperms.

Answer: The life cycle of angiosperms includes seed → germination → plant → flower → pollination → fertilisation → seed and fruit → seed again. (See the diagram in the book.)




Labelled diagram — schematic; parts as named in the answer. See the book diagram for exact form.

Question: 21. Write a detailed note on asexual reproduction methods in plants.

Answer: Asexual reproduction in plants takes place by runners, rhizomes (ginger), tubers (potato), bulbs (onion), suckers, budding, spores and tissue culture; in all of these a new plant is formed from a part of the parent plant without gametes.

Question: 22. Describe asexual reproduction methods in animals.

Answer: Asexual reproduction in animals takes place by binary fission (Amoeba), budding (Hydra), fragmentation (Planaria) and parthenogenesis.

Question: 23. Describe the process of spermatogenesis in detail.

Answer: In spermatogenesis, in the seminiferous tubules of the testes, a primary spermatocyte divides by meiosis to form four haploid spermatids, which mature into sperms (male gametes).

Question: 24. What is pollination, and what are the characteristics of insect-pollinated plants?

Answer: Pollination is the transfer of pollen grains from the anther to the stigma; insect-pollinated plants have colourful, fragrant and nectar-bearing flowers with sticky pollen.

Question: 25. What is seed germination, its requirements, and its different methods?

Answer: Seed germination is the process in which a new plant emerges from the seed; it needs water, suitable temperature and oxygen; its methods are epigeal (seed leaves above the ground) and hypogeal (seed leaves below the ground).

# وراثت

1. The branch of biology studying heredity and variation is called:

(a) inheritance

(b) heredity

✓ (c) genetics

(d) evolution

2. The way parents transfer their genes to offspring is called:

(a) genetics

✓ (b) inheritance

(c) allele

(d) variation

3. In each cell, each chromosome is a thread-like structure whose centre part is called the:

(a) chromatin

(b) arm

(c) homologous

✓ (d) centromere

4. The chemical material of a chromosome is called:

✓ (a) chromatin

(b) chromosome

(c) chromomere

(d) chromonema

5. The outer part of the DNA ladder, made of sugar and phosphate, is called the:

(a) nucleoprotein

✓ (b) upright

(c) rung

(d) phospholipid

6. The small part of DNA carrying hidden instructions in a language, in which information about making a protein is present, is called a:

(a) nucleotide

(b) polynucleotide

✓ (c) gene

(d) exon

7. Copying of DNA is called:

(a) duplication

✓ (b) replication

(c) transcription

(d) transformation

8. If both parents give the same type of trait to a child, it is called:

✓ (a) homozygous

(b) heterozygous

(c) homologous

(d) heterologous

9. A trait's genetic makeup is called the:

✓ (a) genotype

(b) phenotype

(c) genome

(d) phenocopy

Question: 10. What is a gene and how does it work?

Answer: A gene is a part of DNA containing the information for making one protein; through transcription and translation it makes the protein and expresses the body's traits.

Question: 11. Define the terms: homozygous, heterozygous, dominant

allele, allele, recessive allele, homologous, heterologous.

Answer: Homozygous: both alleles alike; heterozygous: both alleles different; dominant allele: the one expressed in the heterozygous condition; allele: different forms of a gene; recessive allele: the one expressed only in the homozygous condition; homologous: alike chromosomes; heterologous: different chromosomes.

Question: 12. Which traits appear in the heterozygous condition in a monohybrid cross?

Answer: In a monohybrid cross the dominant trait appears in the heterozygous condition, such as the tall plant (Tt).

Question: 13. What are the causes in the list of the theory of natural selection?

Answer: The causes in natural selection are: struggle for existence, variation, survival of the fittest, and transfer of traits from generation to generation.

Question: 14. Compare artificial selection with natural selection.

Answer: Natural selection takes place through nature and the fittest organisms survive; artificial selection is done by humans, who select traits of their choice and breed them.

Question: 15. Describe Watson and Crick's model of DNA in detail.

Answer: According to Watson and Crick's model, DNA is a double helix made of two polynucleotide chains; the outer part (upright) is made of sugar and phosphate, the inner rungs are made of nitrogenous bases — adenine always pairs with thymine and cytosine always pairs with guanine by hydrogen bonds.

Question: 16. Describe how a gene expresses itself / how gene expression occurs.

Answer: Gene expression takes place in two stages: in transcription mRNA

is made from DNA, and in translation the information in the mRNA is used to make a protein on the ribosome, which expresses the trait.

Question: 17. Describe Mendel's law of segregation and explain it with a cross.

Answer: According to the law of segregation, the two factors (alleles) of each trait separate from each other when gametes are formed, and each gamete receives one factor; example: in a cross of tall (TT) and short (tt) plants all F1 plants are tall (Tt), and in F2 the ratio of tall to short is 3:1.

Question: 18. Describe Darwin's theory of natural selection with reference to inheritance.

Answer: According to Darwin, variations in organisms are passed to offspring through inheritance; organisms with variations suited to the environment survive and produce more offspring, so their traits increase from generation to generation.

Question: 19. Describe Darwin's theory of natural selection.

Answer: According to Darwin's theory of natural selection, organisms produce more offspring than needed, there is a struggle for existence, variation exists, the fittest survive and pass their traits to offspring, and species change over time.

# انسان ان جو ماحول

1. The layer around the earth that sustains life is called the:

(a) biomass

(b) biome

✓ (c) biosphere

(d) environment

2. Animals of one kind living in one area are called a:

(a) community

(b) species

(c) genus

✓ (d) population

3. An area where a community is in relation with the non-living environment is a:

(a) community

(b) ecology

✓ (c) ecosystem

(d) biome

4. A biological region recognized by particular plants and climate is a:

(a) ecosystem

✓ (b) biome

(c) biosphere

(d) biomass

5. Food material passing from producers among organisms as one eats another is called a:

(a) food pyramid

✓ (b) food chain

(c) food web

(d) ecological pyramid

6. Ecosystem elements recycled through organisms are called the:

(a) food chain

(b) food web

(c) chemical cycle

✓ (d) biogeochemical cycle

7. The process in which nitrogen compounds in organisms change into nitrates is:

(a) ammonification

✓ (b) nitrification

(c) denitrification

(d) deamination

8. The cold war between organisms in an ecosystem is called:

(a) predation

✓ (b) competition

(c) symbiosis

(d) commensalism

9. The mutual relation of two different organisms in which both benefit and neither can live without the other is:

(a) mutualism

✓ (b) symbiosis

(c) commensalism

(d) predation

10. The increase in the amount of harmful gases in the air besides oxygen is:

✓ (a) air pollution

(b) thinning of ozone

(c) acid rain

(d) greenhouse effect

Question: 11. How does light affect the ecosystem?

Answer: Light is essential for photosynthesis, by which plants make food and the whole food chain of the ecosystem runs; lack of light stops growth.

Question: 12. What is the first trophic level and why?

Answer: The first trophic level is the producers (plants), because they make their own food from the sun's energy and feed all other organisms.

Question: 13. What is meant by nitrogen fixation and how does it take place in the ecosystem?

Answer: Nitrogen fixation is the conversion of atmospheric nitrogen into compounds that plants can use; it takes place through lightning and nitrogen-fixing bacteria.

Question: 14. What is called a mutual relationship (symbiosis)?

Answer: Symbiosis is the mutual relationship of two different organisms in which both benefit each other, such as algae and fungi in lichens.

Question: 15. What is called the pyramid of numbers?

Answer: The pyramid of numbers is the chart showing the number of organisms at each trophic level in an ecosystem.

Question: 16. What are called greenhouse effects?

Answer: The greenhouse effect is the trapping of the sun's heat on the earth by gases like carbon dioxide, which raises the earth's temperature.

Question: 17. What is algal bloom and how does it destroy the ecosystem?

Answer: Algal bloom is the rapid growth of algae in water; it uses up the water's oxygen and the water organisms die, which destroys the ecosystem.

Question: 18. What steps can be taken to stop water pollution?

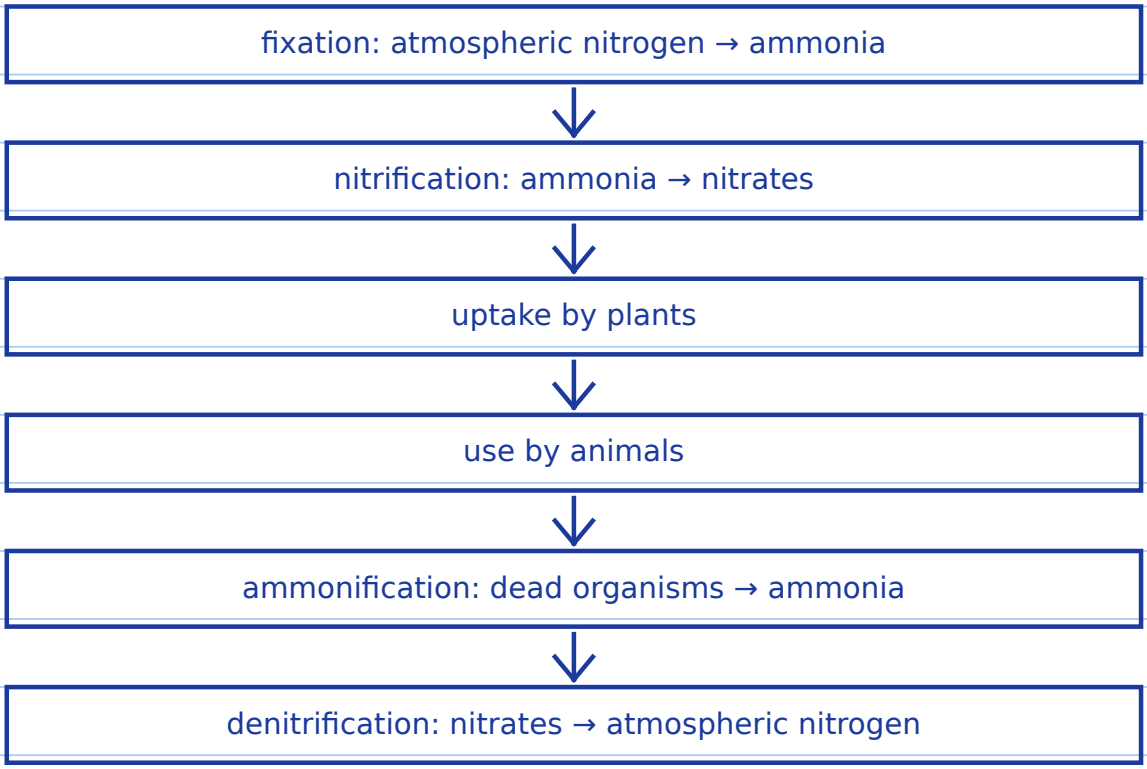
Answer: Factory waste water should be cleaned before releasing into rivers, garbage should not be thrown into water, less fertiliser and pesticide should be used in fields, and water should be saved.

Question: 19. Write names of some endangered species found in Pakistan.

Answer: Endangered species found in Pakistan include the markhor, snow leopard, Indus dolphin, green turtle and houbara bustard (talor).

Question: 20. Draw a diagram of the nitrogen cycle.

Answer: The nitrogen cycle includes fixation (atmospheric nitrogen → ammonia), nitrification (ammonia → nitrates), uptake by plants, use by animals, ammonification (dead organisms → ammonia) and denitrification (nitrates → atmospheric nitrogen). (See the diagram in the book.)



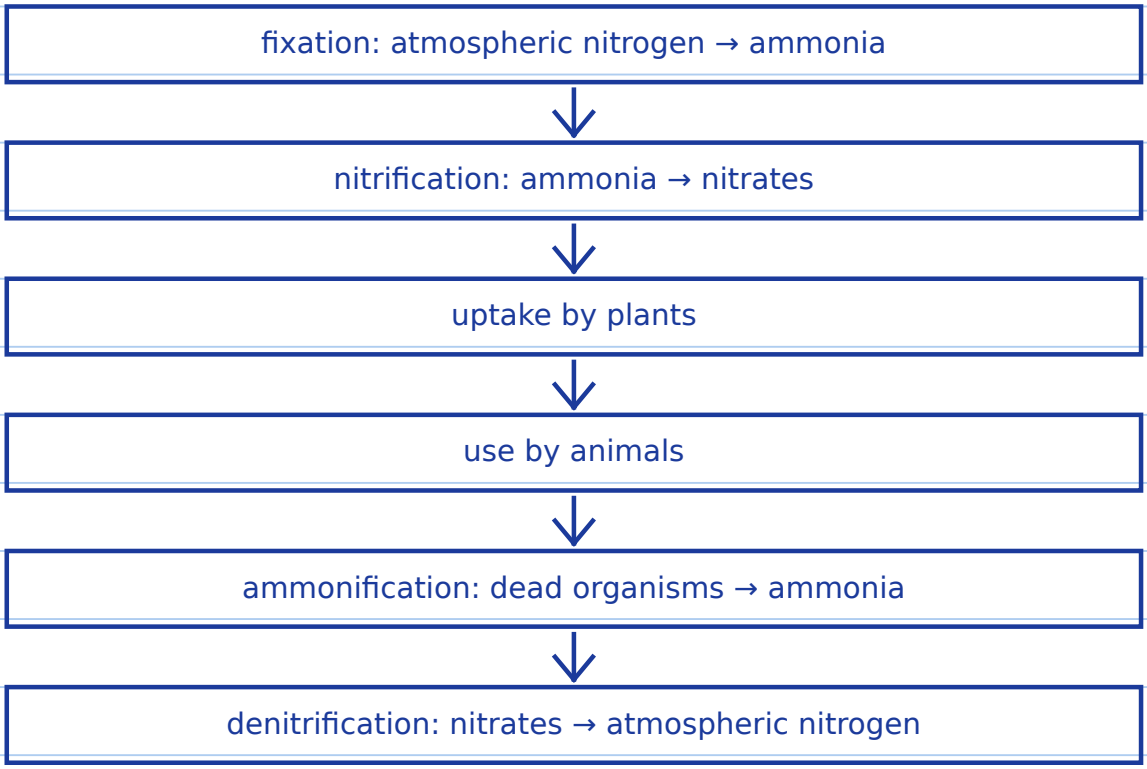

Labelled diagram — schematic; parts as named in the answer. See the book diagram for exact form.

Question: 21. Describe the biotic components of the ecosystem in detail.

Answer: The biotic components of an ecosystem are three: producers (plants which make food), consumers (animals which eat plants or other animals — primary, secondary and tertiary), and decomposers (bacteria and fungi which decay dead organisms and return substances to the soil).

Question: 22. Describe the nitrogen cycle as a biogeochemical cycle.

Answer: In the nitrogen cycle nitrogen circulates between air, soil, plants and animals: fixation changes atmospheric nitrogen into compounds, plants take it up and make protein, animals eat plants, ammonia forms from dead organisms, and denitrification returns nitrogen to the air; thus nitrogen circulates between biotic and abiotic parts.




Labelled diagram — schematic; parts as named in the answer. See the book diagram for exact form.

Question: 23. What is pollution — describe the harmful effects of air pollution in detail.

Answer: Pollution is the addition of harmful substances to the environment; air pollution causes breathing diseases, asthma, bronchitis and lung cancer, irritation of the eyes, acid rain which damages crops, and damage to the ozone layer.

# بائیوٹیکنالاجي

1. Artificially changing/modifying/rearranging DNA is:

✓ (a) genetic engineering

(b) biotechnology

(c) molecular biology

(d) genetics

2. The early biotechnologists were:

(a) biologists

(b) agricultural experts

(c) geneticists

✓ (d) farmers

3. The study of the complete map of the human genome is the:

(a) PCR

✓ (b) HGP

(c) gene therapy

(d) somaclonal

4. Large-scale preparation of alcohol and antibiotics through organisms is a special part of which science:

(a) environmental biotechnology

(b) fermentation

✓ (c) industrial biotechnology

(d) medicinal biotechnology

5. Most organisms use oxygen to produce:

✓ (a) ATP

(b) alcohol

(c) organic acid

(d) ecological pyramid

6. In acidic fermentation, lactic acid is formed from which compound:

✓ (a) pyruvic acid

(b) acetic acid

(c) citric acid

(d) formic acid

7. During alcoholic fermentation the dough rises because of:

(a) methyl alcohol

✓ (b) carbon dioxide

(c) ethyl alcohol

(d) water

8. The vessel used for large-scale production of bacteria is called a:

(a) chiller

(b) sterilizer

✓ (c) fermenter

(d) incubator

9. The naturally occurring enzymes that bacteria use for their defence are:

(a) defence proteins

✓ (b) restriction enzymes

(c) hydrolytic enzymes

(d) ligase enzymes

10. The extra viral DNA used as a vector for a gene is the:

(a) genome

✓ (b) plasmid

(c) chromosome

(d) ligase

Question: 11. Which enzymes are used for cutting and rejoining DNA to make recombinant DNA?

Answer: Restriction enzymes are used for cutting DNA and ligase enzymes for rejoining it.

Question: 12. How do organisms produce energy — how does fermentation work?

Answer: Organisms produce energy (ATP) by breaking down food; in fermentation glucose breaks down without oxygen into pyruvic acid and then into alcohol or lactic acid, giving a little energy.

Question: 13. What kinds of organisms carry out fermentation?

Answer: Yeast, some bacteria (such as Lactobacillus) and the muscles of animals in the absence of oxygen carry out fermentation.

Question: 14. Which food and non-food molecules are produced through fermentation?

Answer: Fermentation produces food and non-food molecules such as curd, cheese, bread, alcohol, vinegar, antibiotics and organic acids.

Question: 15. State the advantages and disadvantages of fermentation in making food usable.

Answer: Advantages: food becomes tasty and easy to digest, keeps longer and gains nutritional value; disadvantages: bad fermentation can spoil food and produce harmful substances.

Question: 16. Which factors affect fermentation?

Answer: Fermentation is affected by temperature, pH, the presence of oxygen, the amount of food (sugar) and the type of organism.

Question: 17. Biotechnology is an ancient technology — prove it.

Answer: For thousands of years humans have made bread, curd, cheese and vinegar by fermentation and selected better crops and animals by breeding; all of these are ancient forms of biotechnology.

Question: 18. Define traditional biotechnology.

Answer: Traditional biotechnology is the preparation of food and other

products using organisms or their parts, such as making bread and curd by fermentation.

Question: 19. Define biotechnology.

Answer: Biotechnology is the preparation of useful products and services for humans using biological systems, organisms or their parts.

Question: 20. Describe Louis Pasteur's services in the development of traditional biotechnology.

Answer: Louis Pasteur proved that fermentation is caused by living organisms (yeast), gave the method of pasteurization, and laid the basis for vaccine preparation.

Question: 21. Describe the different types of fermentation.

Answer: The main types of fermentation are: alcoholic fermentation (alcohol + carbon dioxide), lactic acid fermentation (lactic acid) and acidic fermentation (acids like vinegar).

Question: 22. What is Lactobacillus — what is its role in making yogurt/curd?

Answer: Lactobacillus is a bacterium which converts the sugar of milk (lactose) into lactic acid, so the milk sets into curd.

Question: 23. Which Pakistani foods are made through fermentation?

Answer: Pakistani foods made through fermentation include curd, lassi, pickles, vinegar, leavened bread (naan, double roti) and cheese.

Question: 24. Name 4 products obtained from recombinant DNA available in the market.

Answer: Products obtained from recombinant DNA include insulin, human growth hormone, hepatitis B vaccine and interferon.

# دواسازي

1. Who is considered the founder of antiseptic:

(a) Alexander Fleming

(b) Edward Jenner

✓ (c) Lister

(d) Louis Pasteur

2. The medicine used for treating Rheumatoid Arthritis is obtained from:

(a) animals

(b) minerals

✓ (c) plants

(d) microorganisms

3. Drugs that slow down brain activity are:

(a) narcotics

(b) hallucinogens

(c) marijuana

✓ (d) sedatives

4. A vaccine is given:

(a) after disease

✓ (b) before disease

(c) during disease

(d) in all of these

5. Things that stop the growth of bacteria are:

(a) vaccine

(b) bactericidal

✓ (c) bacteriostatic

(d) antibiotic

6. Haris is a drug addict with effects including seeing imaginary things and sudden happiness — which drug would he be using:

(a) narcotics

✓ (b) hallucinogens

(c) antibiotics

(d) antiseptic

7. Which of the following is caused by misuse of antibiotics:

(a) diarrhoea

(b) immunization

(c) stomach disorder

✓ (d) antibiotic resistance

Question: 8. Why don't antibiotics work against viruses?

Answer: Antibiotics act on the walls and processes of bacteria; viruses lack those structures and live inside cells, so antibiotics do not affect them.

Question: 9. Why is the use of sedative drugs increasing?

Answer: The use of sedative drugs is increasing because of stress, anxiety and sleeplessness; people rely on them for rest and sleep and then become addicted to them.

Question: 10. Why is drug addiction considered dangerous?

Answer: Drug addiction damages the brain and the body, destroys health, creates problems in the family and society, and makes a person a slave of the drug, so it is considered dangerous.

Question: 11. How are medicines obtained from natural sources?

Answer: Medicines from natural sources are obtained by extracting and purifying substances from plants (leaves, bark, roots), animals (glands, liver) and microorganisms (fungi, bacteria).

Question: 12. Can medicines be obtained from animals — if yes, write some examples.

Answer: Yes, medicines can be obtained from animals; examples: insulin (from the pancreas of cow/pig), heparin (from the liver) and cod liver oil (from fish liver).

Question: 13. Does the use of antibiotics cause us any harm — if yes, give some examples.

Answer: Yes, misuse of antibiotics causes harm; examples: destruction of useful bacteria, allergy, diarrhoea and antibiotic resistance.

Question: 14. How does a vaccine work against disease-causing germs — describe this process with a diagram.

Answer: A vaccine contains weakened or dead germs; the body recognises them and makes antibodies and memory cells against them; when the real germs arrive the body destroys them at once. (See the diagram in the book.)

Question: 15. How do bacteria develop resistance against antibiotics?

Answer: In bacteria mutation produces a resistant trait; they make enzymes that break down the antibiotic, pump it out, or change their target; resistant bacteria survive and multiply, and resistance spreads to other bacteria.

Question: 16. Describe the method and working of vaccination.

Answer: In vaccination the vaccine is given into the body by injection or drops; it contains weakened or dead germs which do not cause disease but produce immunity (antibodies and memory cells) in the body, so that the real disease is prevented.